LEVEL 3 · ENTERPRISE / REMOTE ACCESS (DMZ)

FW-01 Industrial firewall
DMZ · read-only data mirror
Only Level 2 ↔ Level 3 conduit

Remote monitoring
Ministry / O&M
VPN + MFA (TBC by client)

VPN

LEVEL 2 · SUPERVISORY – SCADA LAN, VLAN 10, 10.10.10.0/24 (CONTROL ROOM)

HMI-01
Operator station 1
HMIP6 · MSE runtime 4K

HMI-02
Operator station 2
HMIP6 · MSE runtime 4K

SRV-01
SCADA server / historian
13-month storage (TBC)

ENG-01
Engineering station
Control Expert · MSE BT

GPS-01
GPS master clock
NTP / PTP time source

SW1 · MCSESM083F23F1
Managed switch 8TX, control room
VLAN 10 / 20, RSTP, SNMPv3

SCADA LAN – VLAN 10 (via SW1)

LEVEL 1 · CONTROL – M580 PLANT CONTROLLER, VLAN 20, 10.10.20.0/24

NOC0311 #1 / #2

MAIN RACK – BMEXBP0800 (control room)
BMXCPS3020 power supply 24–48 V DC
BMEP582040 M580 CPU – plant controller / PPC
BMENOC0311 #1 – SCADA LAN (Modbus TCP server)
BMENOC0311 #2 – second SCADA LAN path
BMENOC0321 – device network scanner, IP forwarding
BMXDDI3202K ×2 · BMXDDO3202K ×2
Spare BMEP582040 held as cold spare (see note 3)

RIO

RACK 2 – BMEXBP1200 (11 kV GIS room)
BMXCPS3020 power supply
BMECRA31210 Ethernet drop adapter
BMXDDI3202K ×3 – breaker status, trips, fire
BMXDDO3202K ×2 – breaker / DG / BESS commands
BMXNOM0200 – RS485 ×2 (DG controllers)
MCSESM083F23F1 SW2 mounted in this panel

SW2 · MCSESM083F23F1
Device network switch, VLAN 30
Fibre ring to field switches

NOC0321 → SW2 (device network, IP forwarding)

Fibre ring via field switches – see sheet 002

RS485 to DGC-1/2

LEVEL 0 · FIELD DEVICES – DEVICE NETWORK, VLAN 30, 10.10.30.0/24, AND SERIAL BUSES

CONTROL ROOM & 11 kV GIS ROOM

P5M30-AH1 … AH8
8 × Easergy P5M30, RSTP ring
Protection, measurement, events

GIS AH1–AH8 hard-wired
VCB / switch positions, trips, SF6,
L/R, VPI → Rack 2 DI; CLOSE/OPEN DO

NECR (AH8)
Earthing TX trip, NER 63.5 Ω
temperature via P5M30-AH8

UPS-01 + 110 V DC charger
Control room UPS, GIS DC system
Modbus TCP card

FACP-01 fire panel
GIS room, control room zones
Volt-free contacts → Rack 2

AUX panel (AH5)
XRNT-3A fuse, 100 kVA TX, 200 A
Incomer status → Rack 2

BESS CONTAINER

BESS-EMS (vendor)
Energy management controller
Modbus TCP server to M580

PCS-1 / PCS-2
Grid-forming PCS (qty TBC)
via EMS

BMS racks 1–8
3.34 MWh LFP 416S8P
via EMS

PM5320-BESS
BESS LV metering 0.69 kV

BESS ACB / fire / ESD
ACB 1000 A, fire, off-gas,
E-stop → Rack 2 DI / DO

DIESEL GENERATOR AREA

DGC-1 (DG1 550 kVA)
Genset controller
RS485 → BMXNOM0200 port 1

DGC-2 (DG2 660 kVA)
Genset controller
same bus, ID 2

PM5320-DG
DG LV bus metering

ACB Q2 / Q3, start/stop
Hard-wired → Rack 2

PV INVERTER STATION & ARRAY

NTS-PV1 … PV4
4 Edge I/O NTS islands with
NTSFMB0120K serial master each

INV NB01 … NB04
Sungrow SG350HX 350 kW
RS485 → NTSFMB0120K

PAS600-PV1 … PV4
Panel servers + TH110 / CL110
enclosure temperature / humidity

WS-01 weather station
Meteocontrol blue'Log XM 5000
SMP12-A, WindSonic, Tamb, Tmod

PM5320-PV
PV LV metering 0.8 kV

PV ACB 1250 A
Status / command → Rack 2

PRESIDENT TX STATION

NTS-LVP1 / LVP2
RMU, TX, LVDB I/O

PAS600-LVP1 / 2
TH110 / CL110

PM5320-LVP
LVDB metering

RMU ST260E
contacts → NTS

UPS-02 2 kVA
RMU UPS card

PM TX STATION

NTS-LVM1
RMU, TX, LVDB

PAS600-LVM1
TH110 / CL110

PM5320-LVM
LVDB metering

RMU ST260E
contacts → NTS

UPS-03 2 kVA
RMU UPS card

LEGEND

- EtherNet/IP (M580 RIO / DIO)
- Ethernet copper – Modbus TCP / EIP
- Ethernet single-mode fibre (FO)
- RS485 – Modbus RTU
- Hard-wired I/O (DI / DO)
- Vendor-internal link (not SCADA scope)
- Device; stripe = interface type to SCADA
- Network level / location boundary

NOTES

1. Architecture follows the Schneider Electric offer (Villa Somalia v1) and drawing BFE/PR/26/01. Device names are used unchanged on sheets 002–005, the HMI and the I/O list.
2. The M580 is the plant controller (PPC): it dispatches BESS, PV and DG setpoints. Protection relays trip independently of SCADA.
3. BMEP582040 is not hot-standby capable. The second CPU in the BOM is shown as a cold spare. Change to an M580 hot-standby CPU if redundancy is required (TBC).
4. SCADA server, historian, engineering station, GPS clock and firewall are not in the BOM and must be added (TBC).
5. Field switches at remote locations, BESS EMS interface, DG controller interface and FACP interface are not in the BOM (TBC).
6. Only the firewall conduit connects Level 2 to Level 3. Remote access is subject to client approval.
7. Equipment locations and cable distances to be confirmed against the general layout.

REV	DESCRIPTION	DATE
A	Draft for internal review	16-OCT-2026

CLIENT:
MINISTRY OF ENERGY & WATER RESOURCES – SOMALIA
SITE: VILLA SOMALIA

CONTRACTOR:
BLUEFLAG ENERGY CO. LTD

PROJECT TITLE:
SOLARIZATION VILLA SOMALIA

SHEET TITLE:
SCADA ARCHITECTURE
System overview by level and location

SCALE: NTS	DATE: 16-OCT-2026	SIZE: A3
DESIGNED		SIGN
DRAWN		SIGN
CHECKED		SIGN
APPROVED		SIGN

DRW NO. **BFE/PR/26/01-SCADA-001** **REV A**